

Software Bill of Materials (CycloneDX)

Grounded in: Insulet Omnipod 5 (AID patch pump)

Basis: CycloneDX 1.5; NTIA minimum elements.

Device	Insulet Omnipod 5 (controller/app + pod firmware)
Format	CycloneDX 1.5 — machine + human readable; generated in CI
Note	Illustrative structure.

SBOM structure (CycloneDX)

```
{
  "bomFormat": "CycloneDX", "specVersion": "1.5",
  "metadata": {
    "component": { "type": "application", "name": "omnipod5-controller", "version": "" },
    "supplier": { "name": "Insulet Corporation" }
  },
  "components": [
    { "type": "library", "name": "openssl", "version": "3.0.x",
      "licenses": [{ "license": { "id": "Apache-2.0" } } ],
      "properties": [{ "name": "support:level", "value": "supported" } ] },
    { "type": "library", "name": "network-stack", "version": "x.y" },
    { "type": "operating-system", "name": "rtos", "version": "x.y" }
  ],
  "vulnerabilities": []
}
```

What an excellent SBOM adds

- Mobile app + embedded pod components, each with supplier/license/version.
- Support/EOL per component; linked known-vulnerability list; continuous regeneration.

Sources (public)

- Insulet — Omnipod 5, first FDA-cleared tubeless automated insulin delivery (AID) patch pump; smartphone-controlled.
- Public facts: integrated dosing algorithm adjusts insulin every 5 min using CGM input; 510(k) algorithm update (Dec 2025).
- FDA — Premarket Cybersecurity guidance (2025); PCCP final guidance (2024); GMLP.